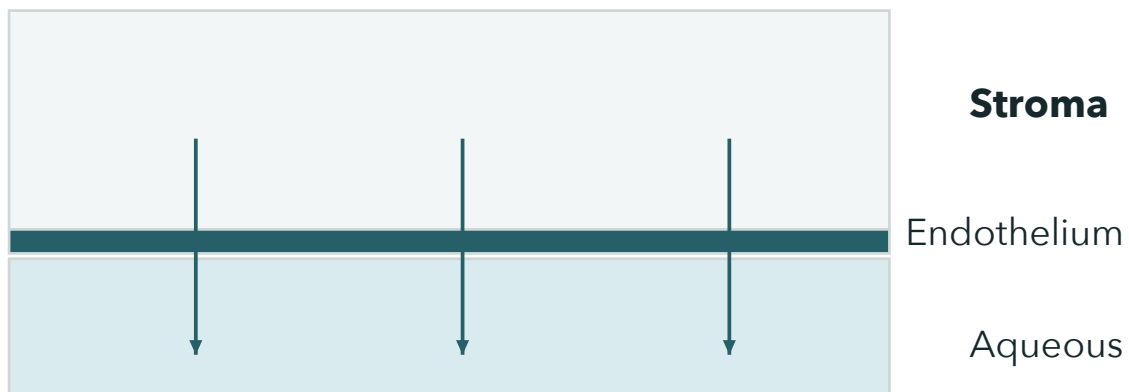


Endothelial grafts: edema, rejection and detachment

Part 2 | Edema is a mechanism, not a diagnosis

Separate pump failure from pressure, inflammation and detachment.

Barrier and pump limit stromal hydration.



Original schematic: a hydrated stroma depends on an intact endothelial barrier and pump.

Assess the time course, IOP, epithelial bullae, keratic precipitates and the anterior chamber. Specular imaging may fail in a cloudy cornea, and a single density value does not summarize function. AS-OCT can reveal posterior-layer separation; pachymetry tracks swelling but cannot by itself identify its cause.

Fuchs changes cataract planning

Assess functional reserve and the likelihood of postoperative decompensation.

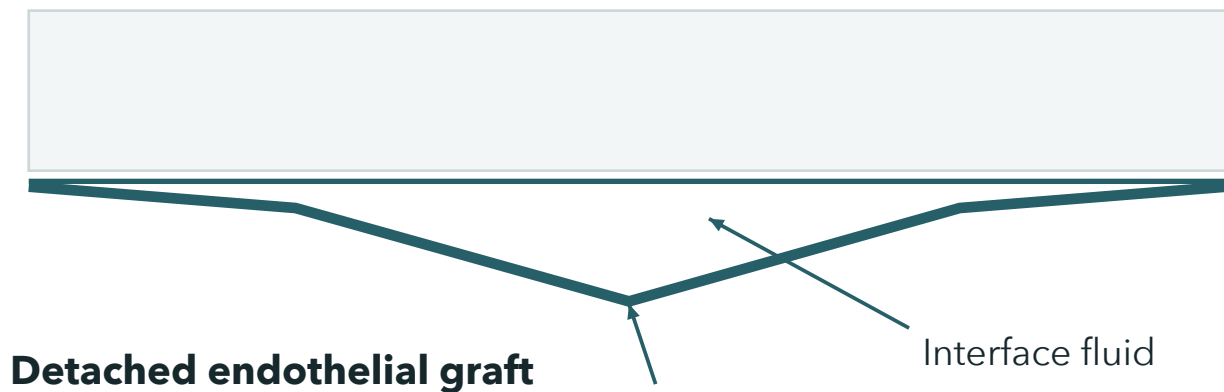
Observation	Planning implication
Morning blur or edema	Pump function is already impaired
Guttae with clear stroma	Reserve still needs individualized assessment
Cell morphology and imaging quality	Do not rely on density alone
Corneal and cataract contributions	Consider staged or combined rehabilitation

Discuss cataract surgery alone versus combined or staged endothelial treatment using symptoms, clinical edema, imaging and surgical demands. Avoid a single universal cell-density threshold. An eye can have relatively preserved counts with abnormal size and shape distribution, or an unmeasurable mosaic because edema obscures it.

A detached DMEK graft is a mechanical problem

Use OCT to distinguish interface separation from unexplained graft edema.

Recipient posterior surface



Original section: central interface fluid prevents graft apposition despite viable tissue.

Small peripheral detachments may behave differently from central, extensive or progressive separation. Rebubbling can restore apposition in selected viable, correctly oriented grafts. Consider location, extent, time course and endothelial function; steroid escalation alone does not mechanically reattach a graft, and immediate full-thickness replacement is not obligatory.

Prior clarity changes the differential

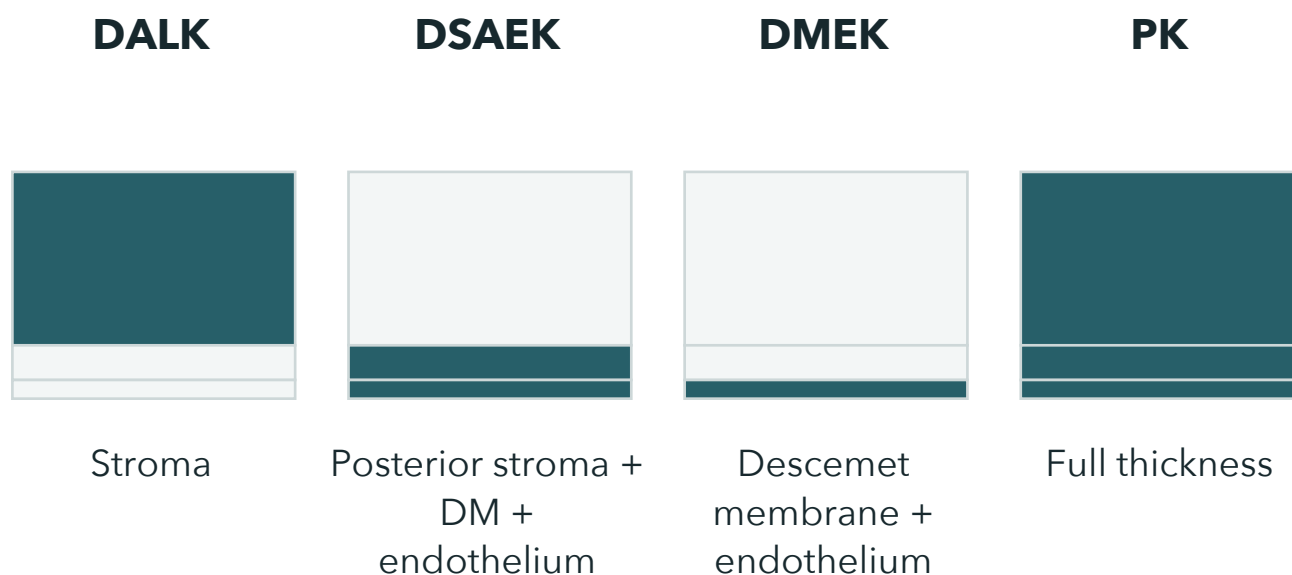
A previously clear graft becoming inflamed is different from one that never functioned.

Course	Important possibility
Never achieves useful clarity	Primary failure or unresolved operative problem
Clear, then edema with new KPs	Immune endothelial rejection
Clear, then edema without inflammation	Late endothelial failure or another cause
Edema with raised IOP	Pressure-related dysfunction may contribute

Inflammatory endothelial signs need prompt specialist assessment and treatment for suspected rejection. Review adherence, infection, IOP and timing.

Replace the diseased layer

Lamellar operations preserve different recipient structures.



Teal = transplanted tissue | thickness schematic

Original layer map; thickness is schematic and not to scale.

DALK replaces anterior stroma while preserving recipient Descemet membrane and endothelium. DSAEK transfers posterior stroma with Descemet membrane and endothelium; DMEK transfers the latter two alone. PK replaces full thickness. Layer selection determines the interface, rejection targets, surgical complexity and remaining host disease.